EX. NO: 11

NETWORK ADDRESS TRANSLATION (NAT) USING CISCO PACKET TRACER

AIM

To implement Network Address Translation (NAT) in Cisco Packet Tracer and allow private devices to communicate with an outside network.

PROCEDURE

1. Add 2 PCs, 1 switch, 1 router and 1 outside router/cloud.
2. Connect the PCs to the switch, the switch to the router inside interface, and the router outside interface to the cloud.
3. Assign private IP addresses to the PCs and the router inside interface, and a public IP address to the outside interface.
4. Configure the router interfaces with ip nat inside and ip nat outside.
5. Configure static NAT: ip nat inside source static <private IP> <public IP>
6. Configure dynamic NAT using a pool and access-list.
7. Ping the outside network from a PC and check the NAT table using show ip nat translations.

CONFIGURATION

```
Router(config)# interface gig0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip nat inside
Router(config-if)# exit
Router(config)# interface gig0/1
Router(config-if)# ip address 200.1.1.1 255.255.255.0
Router(config-if)# ip nat outside
Router(config-if)# exit
Router(config)# ip nat inside source static 192.168.1.10 200.1.1.10
```

OUTPUT

```
Router# show ip nat translations
Pro  Inside global    Inside local
---  200.1.1.10       192.168.1.10
```

RESULT

NAT was successfully configured and private IP addresses were translated to public IP addresses for communication with the outside network.